

CHAPTER 1

INTRODUCTION

The rapid advancements in the Internet of Things (IoT) have paved the way for smart systems that enhance everyday living by combining automation, efficiency, and connectivity. Smart lighting systems are a prime example of IoT applications, addressing the need for energy-efficient solutions and offering enhanced convenience to users. This mini-project focuses on developing a smart lighting system that integrates Wi-Fi connectivity, mobile application control, and real-time notifications to create a versatile and user-friendly solution.

The system utilizes an ESP8266 Wi-Fi module as the core controller, interfacing with a Light Dependent Resistor (LDR) and an LED. The LDR continuously monitors ambient light levels, allowing the system to automate the LED's operation based on real-time conditions. A custom mobile application, developed using MIT App Inventor, enables users to remotely monitor and control the LED, offering flexibility and ease of use.

To further enhance interactivity, the system incorporates a Telegram Bot, which provides real-time notifications about system activity, such as changes in light levels or manual overrides. This feature ensures users are always informed, regardless of their location.

This project addresses the challenges of energy inefficiency and limited user interaction in traditional lighting systems. By automating the control of the LED based on ambient light conditions and enabling remote accessibility, the system enhances convenience and energy conservation. The integration of Telegram notifications further improves user experience by providing instant updates and alerts.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

Existing literature highlights the importance of light-based automation in reducing energy consumption. Systems utilizing Light Dependent Resistors (LDRs) for detecting ambient light levels are commonly employed in applications such as automatic streetlights and home lighting automation. Furthermore, mobile app development platforms like MIT App Inventor provide a user-friendly interface for creating applications that enable remote system control and monitoring.

In this literature survey, we analyze existing works and technologies related to smart lighting systems, IoT integration, and real-time notifications. This review provides a foundation for understanding the current trends and challenges, enabling us to identify gaps and justify the need for our project. The insights gathered will guide the design and implementation of our Wi-Fi-based smart lighting system with MIT App and Telegram Bot integration.

2.2 LITERATURE SURVEY

1. **Innovative Wi-Fi IoT Modules with Ultra-Low Power Wireless Connectivity** By **Vinay Prabhakar Minj** in the magazine **Electronics for You** states that the ESP8266 is widely adopted in smart home applications due to its versatility and support for multiple communication protocols. Studies and examples in IoT projects highlight its reliability in real-time data transfer, making it an integral component in automation systems.

2. **Geethika Kilari; Rizwana Mohammed; Ramesh Jayaraman** published a paper on **Automatic Light Intensity Control using Arduino UNO and LDR** in **IEEE explore** magazine and it tells us that LDRs are effective in applications like automated streetlights and smart lighting systems, offering a cost-efficient method for detecting ambient light.

3. A case study in **DIY Electronics Magazine** demonstrated an LDR-based home lighting system that reduced energy consumption by automatically adjusting lighting levels based on real-time environmental conditions.

4. Mobile apps provide a critical interface for controlling IoT systems. MIT App Inventor, a popular platform for app development, has been highlighted in **Tech Focus Weekly** for enabling beginners to create functional and interactive IoT control applications.

5. Projects documented in **Embedded Systems Design magazine** showcase how MIT App Inventor can be used to design user-friendly apps that facilitate remote monitoring and control of smart devices.

6. **Hack Space Magazine**, illustrated a project where Telegram Bots were used to notify users about environmental changes detected by sensors, demonstrating their effectiveness in enhancing user interaction.

7. The integration of Telegram Bots in IoT projects has gained traction for providing real-time alerts and updates. An article in **IoT Times** described the implementation of Telegram's API for home automation systems, allowing users to receive notifications about device status or anomalies.

CHAPTER 3

PROBLEM STATEMENT AND OBJECTIVES

3.1 PROBLEM STATEMENT

In today's fast-paced world, energy efficiency and real-time monitoring have become critical aspects of smart home and automation systems. Traditional lighting systems often result in energy wastage due to lack of automation and inability to adapt to changing ambient light conditions. Additionally, these systems lack features for remote control and real-time notifications, which are essential for modern, user-friendly solutions.

The absence of an integrated system that combines ambient light detection, automated lighting control, and real-time notifications further exacerbates these challenges. Users need a system that not only adjusts lighting based on environmental conditions but also allows for remote monitoring and manual control via a mobile application, along with instant updates on system activities or anomalies.

This project addresses the following challenges:

1. Reducing energy wastage through automated lighting based on real-time ambient light levels detected by an LDR.
2. Providing remote access and control through a user-friendly mobile app developed using MIT App Inventor.
3. Delivering real-time notifications and alerts via Telegram Bot, ensuring users are informed about system changes and status. The proposed solution leverages the ESP8266 Wi-Fi module for seamless connectivity and data communication, combining automation, interactivity, and energy efficiency into a single smart system.

3.2 OBJECTIVES

1. Design and Development: To design and implement a smart lighting system using the ESP8266 Wi-Fi module, an LDR, and an LED.
2. Ambient Light Detection: To utilize the LDR for real-time measurement of ambient light levels, enabling automated lighting control based on environmental conditions.
3. Automation and Energy Efficiency: To develop an automated system that minimizes energy consumption by controlling the LED based on ambient light intensity.
4. Remote Monitoring and Control: To create a user-friendly mobile application using MIT App Inventor that allows users to remotely monitor system status and manually control the LED.
5. Real-Time Notifications: To integrate a Telegram Bot for delivering real-time notifications and alerts about system status and changes to the user.
6. Seamless Connectivity: To ensure efficient communication and data transfer between the components of the system using the ESP8266 Wi-Fi module.
7. Practical IoT Application: To demonstrate the practical application of IoT concepts in creating a functional, efficient, and interactive smart home solution.
8. User Convenience: To enhance user experience by combining automation, real-time notifications, and remote accessibility in a single integrated system.

CHAPTER 4

METHODOLOGY

4.1 COMPONENTS REQUIRED

1. ESP8266 wi-fi module: The ESP8266 Wi-Fi module is a low-cost, compact microcontroller-based module widely used in IoT (Internet of Things) projects. It offers built-in Wi-Fi connectivity, making it ideal for integrating devices with wireless networks as shown in figure 4.1.1. An ESP8266 Wi-Fi module is a SOC microchip mainly used for the development of end-point IoT (Internet of things) applications. It is referred to as a standalone wireless transceiver, available at a very low price. It is used to enable the internet connection to various applications of embedded systems. ESP8266 Wi-Fi module to support both the TCP/IP capability and the microcontroller access to any Wi-Fi network. It provides the solutions to meet the requirements of industries of IoT such as cost, power, performance, and design.

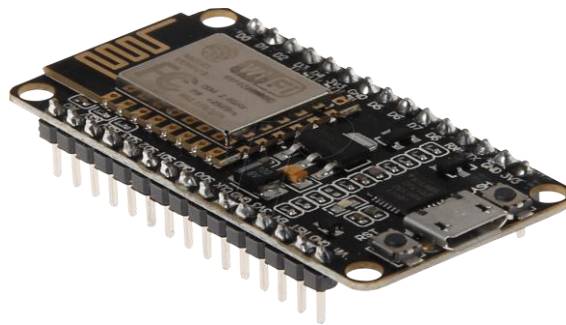


Figure 4.1.1 ESP8266 wifi module

2. LDR(Light-dependent resistor) sensor: An electronic component like LDR or light-dependent resistor is responsive to light. Once light rays drop on it, then immediately the resistance will be changed. The resistance values of an LDR may change over several orders of magnitude. The resistance value will be dropped when the light level increases. The resistance values of LDR in darkness are several megaohms whereas in bright light it will be dropped to hundred ohms. So due to this change in resistance, these resistors are extremely used in different applications. The LDR sensitivity also changes through the incident light's wavelength.

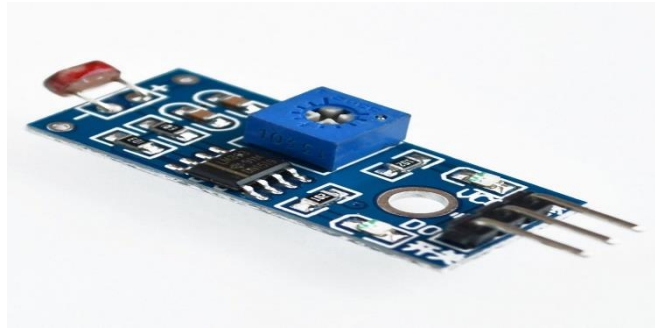


Figure 4.1.2 LDR sensor

4. **LED:** A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it. When current passes through an LED, the electrons recombine with holes emitting light in the process. LEDs allow the current to flow in the forward direction and blocks the current in the reverse direction. Light-emitting diodes are heavily doped p-n junctions. Based on the semiconductor material used and the amount of doping, an LED will emit coloured light at a particular spectral wavelength when forward biased. As shown in the figure, an LED is encapsulated with a transparent cover so that emitted light can come out.



Figure 4.1.3 LED

4.2 SOFTWARE TOOLS USED

1. Arduino IDE 1.8.19: Arduino IDE 1.8.19 is a version of the popular Integrated Development Environment (IDE) used for programming Arduino microcontroller boards. This version includes several updates and fixes aimed at improving stability, usability, and performance.

2. MIT App Inventor: MIT App Inventor is a web-based, user-friendly platform designed to help people create mobile applications for Android devices without requiring advanced programming skills. Developed by the Massachusetts Institute of Technology (MIT), it allows users to build apps using a drag-and-drop interface, which simplifies the app development process.

3. TELEGRAM BOT: A Telegram Bot is an automated program that runs on the Telegram messaging platform, capable of performing specific tasks, interacting with users, and integrating with other systems. Bots can be programmed to send messages, provide information, handle commands, and interact with external APIs, making them highly versatile for a wide range of applications. Bots can perform tasks automatically, such as sending notifications or responding to user inputs. They enable instant communication with users through the Telegram app. Telegram provides a Bot API and Bot Father tool to simplify bot creation and management. Bots can integrate with IoT systems, databases, or third-party services to expand functionality. Developers can program bots to handle specific commands, events, or workflows tailored to the application's needs.

4.3 CIRCUIT DESIGN AND IMPLEMENTATION

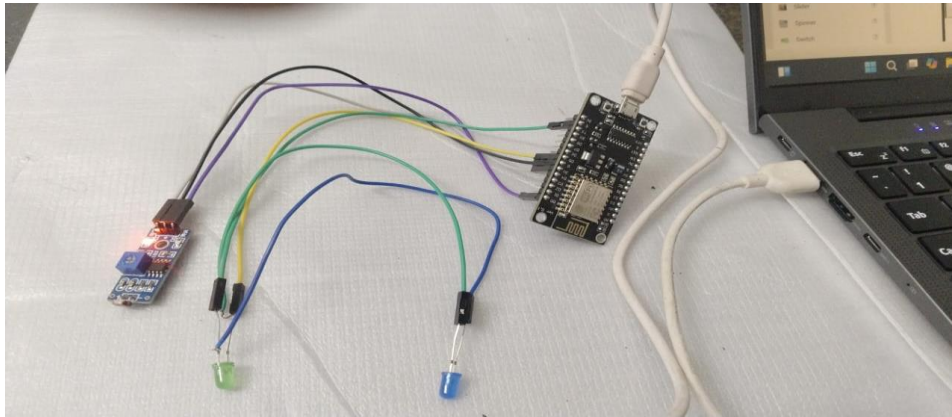


Figure 4.3.1 Hardware circuit design

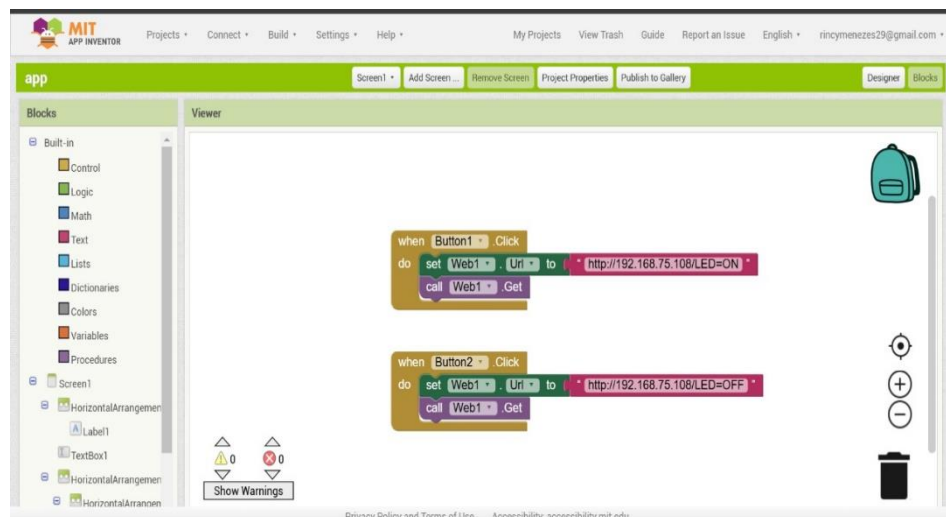


Figure 4.3.2 MIT app inventor back end app design

The LDR (Light Dependent Resistor) measures ambient light intensity. Based on the light level, the system determines whether the LED needs to be turned on or off. The ESP8266 serves as the central controller, receiving data from the LDR and controlling the LED accordingly as shown in the figure 4.3.1. It establishes a Wi-Fi connection to enable remote communication with the mobile app and Telegram bot as shown in figure 4.3.2. The LED acts as the lighting element in the system. It is controlled based on LDR readings and can also be toggled manually through the mobile app. A custom mobile app is developed using MIT App Inventor. The app allows users to monitor light levels, control the LED manually, and view the system status remotely. The system sends real-time notifications to the user via a Telegram Bot. The system operates automatically based on LDR readings but allows users to override automation using the mobile app.

4.4 CODE FOR THE PROTOTYPE

```
#include <ESP8266WiFi.h>
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>
#include <ArduinoJson.h>
WiFiServer server(80);
const int ledPin = D5;
// Replace with your network credentials
#define WIFI_SSID "commando"
#define WIFI_PASSWORD " "

// Telegram BOT Token (Get from Botfather)
#define BOT_TOKEN
"7719149296:AAGy5GkhtFPtn7gqxyKOmiFFAU6X0roRLJ0q"
#define CHAT_ID "7815411476"

// Pin where the LDR is connected
#define LDR_PIN A0

X509List cert(TELEGRAM_CERTIFICATE_ROOT);
WiFiClientSecure secured_client;
UniversalTelegramBot bot(BOT_TOKEN, secured_client);

void setup() {
  Serial.begin(115200);
  Serial.println();
  pinMode(ledPin, OUTPUT);
  // Connect to Wi-Fi
  Serial.print("Connecting to Wifi SSID ");
  Serial.println(WIFI_SSID);
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  secured_client.setTrustAnchors(&cert); // Add root certificate for api.telegram.org

  while (WiFi.status() != WL_CONNECTED) {
    Serial.print(".");
    delay(500);
  }
  Serial.println("\nWiFi connected. IP address: " + WiFi.localIP().toString());

  // Synchronize time with NTP server
  Serial.print("Retrieving time: ");
  configTime(0, 0, "pool.ntp.org");
  time_t now = time(nullptr);
  while (now < 24 * 3600) {
    Serial.print(".");
```

```
    delay(100);
    now = time(nullptr);
}
Serial.println("Time retrieved: " + String(now));

// Notify Telegram Bot of startup
bot.sendMessage(CHAT_ID, "Bot started up", "");
server.begin();
}

void loop() {
    // Read LDR value
    WiFiClient client = server.available();
    if (client) {
        Serial.println("Client connected!");
        String request = client.readStringUntil('\r');
        Serial.print("Request: ");
        Serial.println(request);
        client.flush();
        if (request.indexOf("LED=ON") != -1) {
            digitalWrite(ledPin, HIGH);
            client.print("HTTP/1.1 200 OK\r\nContent-Type: text/html\r\n\r\nLED is ON");
        }
        if (request.indexOf("LED=OFF") != -1) {
            digitalWrite(ledPin, LOW);
            client.print("HTTP/1.1 200 OK\r\nContent-Type: text/html\r\n\r\nLED is OFF");
        }
        client.stop();
        Serial.println("Client disconnected.");
        int ldrValue = analogRead(LDR_PIN);
        Serial.println("LDR Value: " + String(ldrValue));

        // Check if LDR value exceeds the threshold
        if (ldrValue > 20) {
            Serial.println("LDR value exceeds threshold. Sending Telegram message...");
            bot.sendMessage(CHAT_ID, "Alert! LDR value is above 20: " + String(ldrValue),
            "");
            delay(1000); // Avoid spamming messages (10 seconds delay)
        }

        delay(1000); // Main loop delay
    }
}
```

CHAPTER 5

RESULT

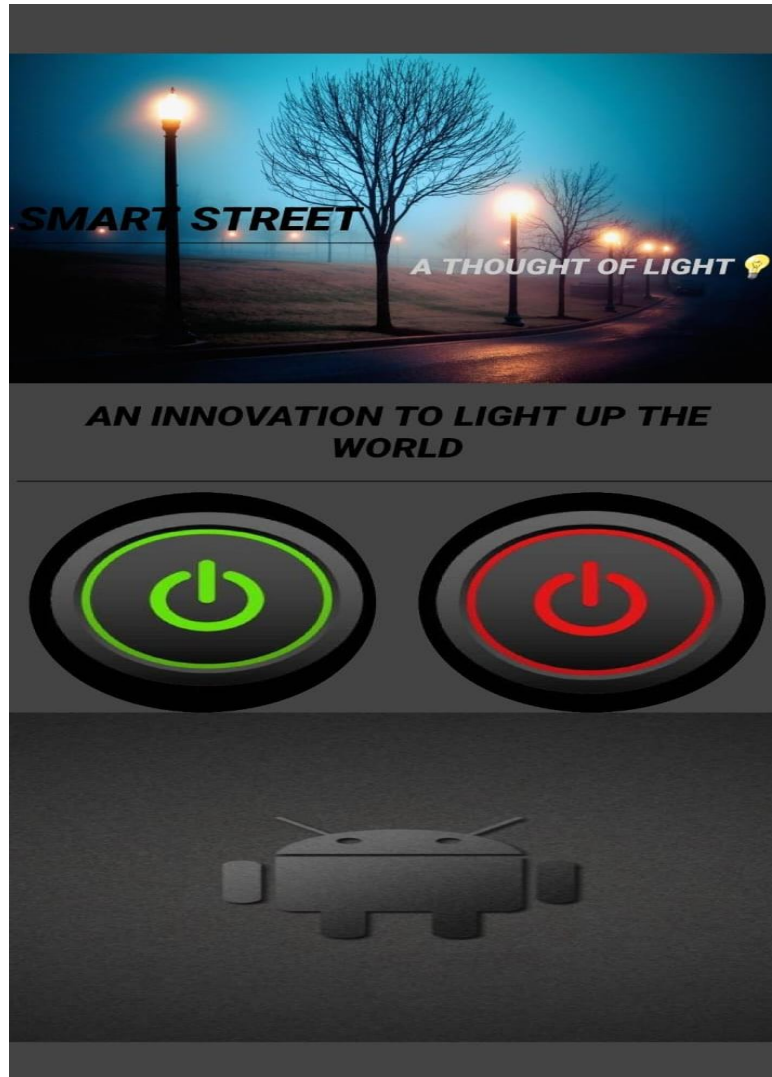


Figure 5.1 User interface of the Mobile app

The implementation of the IoT-based smart lighting system was successful, and the following results were achieved:

The LDR sensor accurately detected ambient light levels and provided real-time data to the ESP8266 Wi-Fi module for processing. Automated LED control was effectively achieved based on predefined light intensity thresholds. The system demonstrated efficient lighting automation by turning the LED on or off depending on the ambient light levels, reducing manual intervention and ensuring energy efficiency. The mobile application developed using

MIT App Inventor successfully provided real-time light level monitoring. Users were able to remotely control the LED through the app, switching between automatic and manual modes seamlessly. The **Telegram Bot** was successfully integrated to deliver instant notifications to the user about system events, such as light intensity changes, LED state updates, and manual overrides as shown in the figure 5.1. Notifications were timely, accurate, and provided meaningful information, enhancing user interaction and system transparency. The **ESP8266 Wi-Fi module** maintained a stable connection to the local Wi-Fi network, enabling seamless communication between the hardware, mobile app, and Telegram Bot. By automating lighting control based on LDR readings, the system minimized unnecessary energy consumption, demonstrating the potential for practical energy-saving applications.

Summary of Results

The system met the expected objectives and successfully integrated hardware components with software interfaces. The combination of automation, real-time notifications, and remote accessibility resulted in a robust, user-friendly IoT solution for smart lighting control. This project validates the feasibility of using IoT to create efficient, interactive, and energy-conscious smart home systems.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

In conclusion, the IoT-based smart lighting system, integrating the ESP8266 Wi-Fi module, LDR sensor, LED, MIT App Inventor, and Telegram Bot notifications, successfully meets the objectives of automation, remote control, and real-time updates. The system effectively detects ambient light levels and automates lighting control, reducing energy consumption by adjusting the LED based on environmental conditions.

The mobile application developed using MIT App Inventor provides a user-friendly interface for monitoring light levels and controlling the system remotely, while the Telegram Bot ensures that users are notified instantly of any system changes or updates. This integration of automation and real-time communication enhances the overall user experience by providing both convenience and control over the lighting system.

This project demonstrates the practical application of IoT concepts in smart home systems and highlights the potential for combining simple, low-cost components to create effective solutions for everyday problems. The system's successful implementation shows that it is possible to create efficient, interactive, and user-friendly smart systems with minimal resources.

While the current project demonstrates the integration of the ESP8266 Wi-Fi module, LDR sensor, LED, MIT App Inventor, and Telegram Bot for smart lighting control, there are several areas for further enhancement and expansion:

1. **Integration with Voice Assistants:** The system could be extended to support integration with voice assistants like Amazon Alexa, Google Assistant, or Siri, enabling users to control the lighting system through voice commands for greater convenience.
2. **Advanced Light Control and Customization:** The LDR sensor can be augmented with multiple sensors to detect different light conditions or add features like dimming the LED based on ambient light intensity, allowing more nuanced control of lighting. The app could include advanced settings to allow users to set specific brightness thresholds or schedules for automatic lighting based on the time of day or their preferences.

3. **Multiple Device Integration:** The system could be expanded to control multiple LEDs or other smart devices simultaneously, enabling the creation of a full home automation system. This would allow users to manage all connected devices from a single mobile application.
4. **Cloud Integration for Data Storage:** By integrating cloud services such as Google Firebase or AWS, data from sensors and device status could be stored and analyzed, providing insights into energy consumption patterns and usage history. This would allow users to track and optimize their energy consumption over time.
5. **Smart Energy Management:** Advanced algorithms could be introduced to optimize energy usage based on user behaviour, weather conditions, or time of day. The system could adjust lighting in real-time based on factors such as sunlight intensity or household activity.
6. **Security and Monitoring:** The system could be extended to include motion sensors or security cameras, providing enhanced home security. The Telegram Bot could send alerts when unusual activity is detected, integrating both smart lighting and security features into one cohesive system.
7. **Mobile App Enhancements:** Additional features could be added to the MIT App Inventor application, such as push notifications for specific events, detailed statistics on energy savings, or remote control of other smart home devices, making the app a more comprehensive home automation tool.
8. **AI and Machine Learning Integration:** The system could incorporate machine learning algorithms to learn user preferences over time and adjust the lighting automatically without user input. For example, the system could learn when the user typically adjusts the lighting and automate this process based on patterns.

By incorporating these features, the system could evolve from a simple lighting control system into a comprehensive smart home solution that provides users with greater convenience, efficiency, and control. The future scope for this project offers a wide range of possibilities for further advancements in automation, smart home technologies, and IoT applications.

REFERENCES

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